

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Course Name: CSA16

Course Code: Data Warehousing and Data Mining

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Course Outcome Covered: CO3 – Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)

Assessment Tool Weightage: 50% **Questions:** 5 **Total Marks:** 50

Annexure A – Index

Criteria	Understanding of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total (10)
Weightage	25%	30%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct

I **O.Siva Charan (192525039)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: O.Siva Charan

Answers

Q1. Chi-Square Test — Age vs Photograph Preference (10 Marks)

Data

Age of child	A	B	C	Row Total
5–6 years	18	22	20	60
7–8 years	2	28	40	70
9–10 years	20	10	40	70
Col Total	40	60	100	200

Step 1 — Expected counts $E = (\text{row total} \times \text{col total}) / \text{grand total}$

Age	A	B	C
5–6	12.0	18.0	30.0
7–8	14.0	21.0	35.0
9–10	14.0	21.0	35.0

Step 2 — Chi-square statistic $\Sigma(O-E)^2/E$

5–6 row: $3.00 + 0.89 + 3.33 = 7.22$

7–8 row: $10.29 + 2.33 + 0.71 = 13.33$

9–10 row: $2.57 + 5.76 + 0.71 = 9.05$

$\chi^2 \approx 29.60$, $df = (3-1)(3-1) = 4$, critical value ($\alpha = 0.05$) = 9.49

Since $29.60 > 9.49 \rightarrow$ **reject H_0** . There is a statistically significant relationship between a child's age and photograph preference.

What is wrong with the study?

- The 7–8 years / Photo A cell has only 2 observations, well below the 'expected ≥ 5 ' rule of thumb for chi-square validity — makes the test unreliable in that cell.
- No description of random sampling, sample selection, or how 'preference' was measured (forced choice vs. rating).
- Only 3 arbitrary photographs/ages — small, non-representative stimulus set limits generalizability.
- Association $\neq$ causation — the test shows correlation between age group and preference, not that age causes the preference.
- Unequal group sizes and no control for confounds (gender, culture, photo ordering effects).

R code and results

```
photo <- data.frame(A = c(18, 2, 20), B = c(22, 28, 10), C = c(20, 40, 40))

cov(photo$B, photo$C) # sample covariance B & C ~= -20.00
cov(photo) # covariance matrix
cor(photo$B, photo$C) # sample correlation B & C ~= -0.189
cor(photo) # correlation matrix
```

Covariance matrix

	A	B	C
A	97.33	-74.00	-46.67

B	-74.00	84.00	-20.00
C	-46.67	-20.00	133.33

Correlation matrix

	A	B	C
A	1.00	-0.82	-0.41
B	-0.82	1.00	-0.19
C	-0.41	-0.19	1.00

Q2. Box Plot — Tennis Player Points (10 Marks)

Sorted data (n = 15): 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, **35**

Median = 8th value = **17**

Q1 = median of lower 7 (10–16) = **13**

Q3 = median of upper 7 (18–35) = **21**

IQR = 21 – 13 = **8**

Lower fence = 13 – 1.5(8) = **1**

Upper fence = 21 + 1.5(8) = **33**

Decision process: any point below 1 or above 33 is flagged an outlier. Only **Player J (35 points)** exceeds the upper fence → it is a statistical **outlier**.

Box plot construction: box from Q1(13) to Q3(21) with a line at median(17); whiskers extend to the smallest/largest non-outlier values (10 and 23); Player J's 35 is plotted as an isolated point beyond the upper whisker.

Q3. Decision Tree — Loan Approval (10 Marks)

Total: 15 records → 9 Yes, 6 No → **Entropy(S) = 0.971**

Information gain per attribute

Attribute	Info Gain
Age	0.083
Employment Status	0.470
Income	0.711
Credit Score (highest)	0.730

Root node = Credit Score

- Credit = Good (5 records) → all Yes → leaf: Yes
- Credit = Poor (5 records) → all No → leaf: No
- Credit = Average (5 records: mixed 4 Yes / 1 No) → split further

Within the Average branch, splitting on **Age** gives a perfect (zero-entropy) split:

Age = Young → Yes | Age = Middle → Yes | Age = Old → No

Final Decision Tree

```
Credit Score?
|-- Good -> Yes
|-- Poor -> No
```

```

`-- Average -> Age?
|-- Young -> Yes
|-- Middle -> Yes
`-- Old -> No

```

Q4. Class A vs Class B — Mean, Median, Range, Box Plot (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84 → sorted: 35,36,47,64,66,76,84,89,95

Mean = $592 / 9 = 65.78$ | Median (5th value) = **66** | Range = $95 - 35 = 60$ | Q1 = 41.5, Q3 = 86.5, IQR = 45

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50 → sorted: 50,51,56,59,60,63,66,70,84

Mean = $559 / 9 = 62.11$ | Median (5th value) = **60** | Range = $84 - 50 = 34$ | Q1 = 53.5, Q3 = 68, IQR = 14.5

(i) Comparison: Class A has the higher mean, higher median, and larger range than Class B.

(ii) Inference from box plots: Class A's box is much wider (IQR 45 vs 14.5) — its scores are more spread out/inconsistent, though its typical (median) score is higher. Class B is tightly clustered around its median, indicating more consistent performance across students but a lower overall level. Neither class has outliers by the $1.5 \times \text{IQR}$ fence rule.

Q5. Age vs %Body Fat — 18 Adults (10 Marks)

Standard dataset used for this question — Age: 23,23,27,27,39,41,45,49,50,53,53,54,56,57,58,58,60,61 | %Fat: 9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7

(a) Mean, median, standard deviation

	Age	%Fat
Mean	46.33	28.78
Median	51.5	30.7
Std. Dev.	~13.22	~9.25

(b) Box plots

Age: Q1 = 39, Median = 51.5, Q3 = 57, IQR = 18

%Fat: Q1 = 26.5, Median = 30.7, Q3 = 34.1, IQR = 7.6

No values fall outside the $1.5 \times \text{IQR}$ fences for either variable → no outliers.

(c) Scatter plot & Q-Q plot

Scatter plot (Age on x-axis, %Fat on y-axis) shows a clear positive linear relationship — as age increases, body fat percentage tends to increase.

Q-Q plot (sorted Age quantiles vs sorted %Fat quantiles) checks whether the two variables follow a similar distribution shape; for this dataset the points lie roughly along a straight line, supporting the linear association seen in the scatter plot.